D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Discuss the development and clinical application of next-generation targeted therapies, including KRAS inhibitors, PARP inhibitors, and antibody–drug conjugates.

January 2026 | DM 0126 | Q1 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction — the ‘next-generation targeted’ triad

This January 2026 essay expects an integrated discussion of three transformative classes: (1) direct KRAS inhibitors (finally drugging the ‘undruggable’), (2) PARP inhibitors exploiting synthetic lethality in homologous recombination deficiency, and (3) antibody–drug conjugates as targeted cytotoxics. Organize by class: biology → drugs/doses → tumor types → pivotal trials → resistance → NCCN 2026 refinements → shared sequencing principles.

EXAM PEARLS / HALLMARKS

- Announce a three-part structure in sentence one so the examiner can follow.

2. KRAS biology — why G12C was first

KRAS cycles between GTP-on and GDP-off states; oncogenic mutations lock signaling through MAPK/PI3K. Codon 12/13/61 alleles differ by tumor: G12C common in NSCLC (smoking-related), G12D dominant in PDAC and common in CRC, G12V across tumors.

Covalent G12C inhibitors bind the mutant cysteine in the switch-II pocket in the GDP state (inactive), trapping KRASG12C. Allele-specific chemistry explains why G12D needs non-covalent / salt-bridge strategies (emerging inhibitors).

KRAS mutant	Chronic GTP signaling
G12C covalent Tx	Switch-II pocket + Cys12 bond
Downstream	RAF–MEK–ERK / PI3K–AKT
Resistance	RTK feedback, new KRAS mutations, fusions
Combos	Anti-EGFR (CRC), SHP2, MEK, IO, chemo

Figure: mechanism / pathway map

3. Sotorasib & adagrasib — approved G12C inhibitors

Sotorasib (CodeBreaK program): oral KRASG12C inhibitor approved for previously treated KRAS G12C NSCLC; CRC activity improved when combined with EGFR inhibition (CodeBreaK 101/301 paradigms) because EGFR feedback reactivates wild-type RAS isoforms in CRC.

Adagrasib (KRYSTAL program): alternative G12C inhibitor with CNS activity signals; KRYSTAL-1 NSCLC; KRYSTAL-1/-12 /-10 combinations with cetuximab in CRC. Choose based on access, interactions, and CNS disease — both are subsequent-line standards pending evolving 1L combinations.

- Toxicities: diarrhea, hepatotoxicity, ILD/pneumonitis rare, QTc (adagrasib monitoring), edema.
- Avoid strong CYP inducers/inhibitors per label; acid-reducing caveats as applicable.

Trial	Key result / exam takeaway
CodeBreaK 100/200	Sotorasib in KRAS G12C NSCLC
KRYSTAL-1	Adagrasib NSCLC ± CNS activity
CodeBreaK / KRYSTAL CRC combos	G12Ci + EGFR mAb improves CRC control

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Sotorasib	960 mg PO daily (NSCLC label dose; modify for interactions/toxicity)
Adagrasib	600 mg PO BID
Adagrasib + cetuximab (CRC)	Per KRYSTAL combination protocols

Table: doses and regimens

4. Emerging KRAS G12D and pan-RAS strategies

G12D inhibitors (e.g., investigational allele-selective agents such as MRTX1133-class / newer clinical candidates) and pan-KRAS/RAS approaches are 2026 exam ‘future directions’. PDAC — dominated by G12D — is the unmet-need poster child.

Combination logic: SHP2 inhibitors, SOS1 inhibitors, MEK inhibitors, immunotherapy, and chemotherapy backbones aim to block adaptive feedback. Mention resistance via secondary KRAS alterations, MET/EGFR activation, and histologic transformation analogies.

- Always re-profile at progression (tissue/ctDNA) after KRAS G12Ci.
- G12C CRC: do not use G12Ci monotherapy as enthusiastically as in NSCLC — pair EGFR blockade.

KRAS allele-directed flow

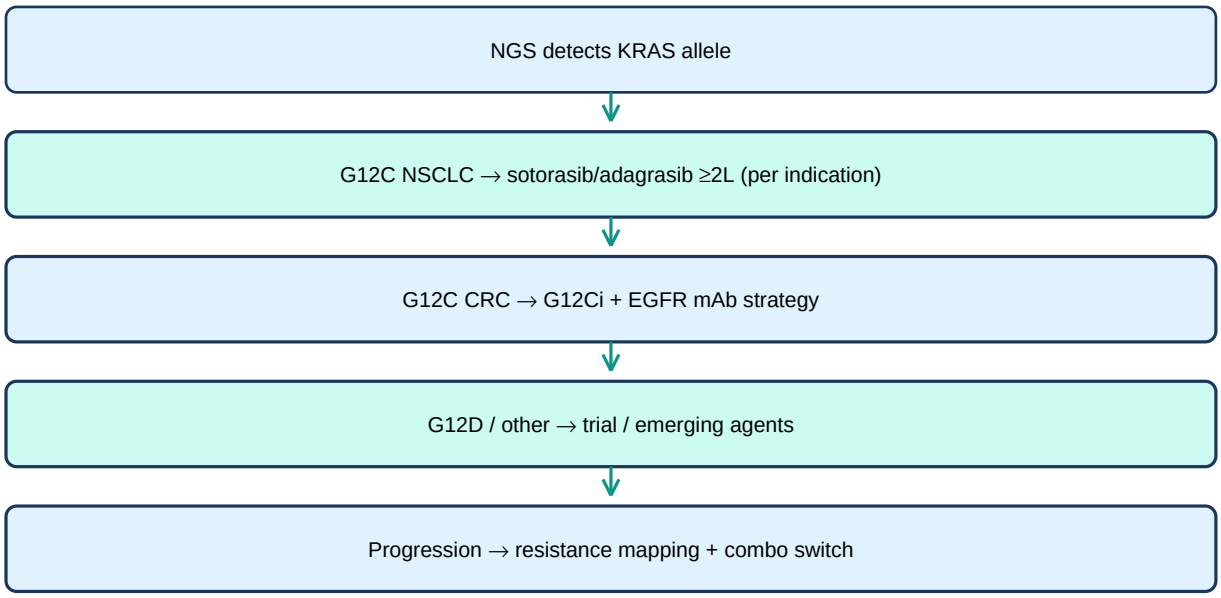


Figure: stepwise exam algorithm

5. PARP inhibitors — mechanism of synthetic lethality

PARP enzymes repair single-strand DNA breaks. PARP inhibitors trap PARP on DNA and block BER repair; in homologous recombination–deficient cells (BRCA1/2, other HRR genes), double-strand break repair fails → synthetic lethality. HRD assays and BRCA status select benefit; platinum sensitivity often correlates.

SSB	PARP recruited
PARPi	Catalytic block + PARP trapping
Replication fork collapse	DSB accumulates
HR proficient	Repair possible → less kill
HR deficient	Synthetic lethality → apoptosis

Figure: mechanism / pathway map

6. PARPi across tumor types

Ovarian: SOLO1 (olaparib maintenance gBRCA after platinum), PAOLA-1 (olaparib+bevacizumab HRD), PRIMA (niraparib broad 1L maintenance), NOVA historical recurrent — but NCCN 2026 narrows PARPi in recurrent treatment sections toward maintenance/biomarker-defined use.

Breast: OlympiAD (olaparib), EMBRACA (talazoparib) in gBRCA mBC; OlympiA adjuvant olaparib for high-risk gBRCA HER2– early breast cancer.

Pancreas: POLO — olaparib maintenance after platinum in gBRCA metastatic PDAC. Prostate: PROfound (olaparib HRR), MAGNITUDE/PROpel/TALAPRO combinations with ARPIs; NCCN prostate update highlights ADT + niraparib/abiraterone for BRCA2-mutated APMS subsets.

Trial	Key result / exam takeaway
SOLO1 / PAOLA-1 / PRIMA	1L ovarian maintenance paradigms
OlympiA / OlympiAD	Adjuvant & metastatic gBRCA breast

POLO	PDAC gBRCA maintenance
PROfound / MAGNITUDE / PROpel	Prostate HRR/BRCA PARPi ± ARPI

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Olaparib tablets	300 mg PO BID
Niraparib	200–300 mg PO daily (individualize by weight/platelets in ovarian)
Talazoparib	1 mg PO daily (breast)
Rucaparib	600 mg PO BID (indication-dependent)
Niraparib + abiraterone	Per prostate combination label with ADT + prednisone

Table: doses and regimens

7. NCCN 2026 PARP refinements — ovarian narrowing & prostate BRCA2

High-yield 2026 points: (1) Ovarian — PARPi removed from many recurrent therapy listings; concentrate on maintenance and biomarker-defined (BRCA/HRD) settings; platinum-resistant non-PARP preferred options include nab-paclitaxel+relacorilant (ROSELLA). (2) Prostate — ADT + niraparib/abiraterone for BRCA2-mutated high-volume (and Cat 2B low-volume) synchronous/metachronous APMS disease. (3) Breast — continue olaparib/talazoparib pathways including OlympiA.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Ovarian Cancer 2026 panel updates; Prostate Cancer v1.2027; Breast Cancer v9.2026
- Ovarian: PARPi narrowed in recurrent treatment sections; focus maintenance + BRCA/HRD.
- Prostate: ADT + niraparib/abiraterone for BRCA2 APMS subsets.
- Breast: olaparib/talazoparib metastatic gBRCA; adjuvant olaparib OlympiA population.
- Counsel MDS/AML rare risk and hematologic monitoring across indications.

EXAM PEARLS / HALLMARKS

- Saying ‘PARPi narrowed in ovarian 2026’ is a discriminator vs outdated answers dumping PARPi on all platinum-resistant disease.

8. PARPi toxicities & resistance

Class toxicities: anemia, neutropenia, thrombocytopenia, nausea, fatigue; niraparib — early hypertension/platelet dips (individualized starting dose). Rare MDS/AML — informed consent. Drug interactions via CYP3A for olaparib.

Resistance: BRCA reversion mutations restoring HR, loss of PARP trapping, drug efflux, replication fork protection pathway changes. Reversion detectable on ctDNA — elegant exam pearl.

- Do not stack full-dose myelotoxic chemo concurrently without protocol.
- Hold PARPi for grade ≥3 hematologic toxicity per label.

9. ADC catalogue within the triad essay

Reprise ADCs concisely (full detail was Feb 2025 Q373): T-DXd (HER2), T-DM1, SG & Dato-DXd (TROP2), EV (Nectin-4), mirvetuximab (FRα), heme ADCs (BV, pola, belantamab, loncastuximab). Emphasize bystander biology and NCCN 2026 expansion.

ADC arm inside the triad

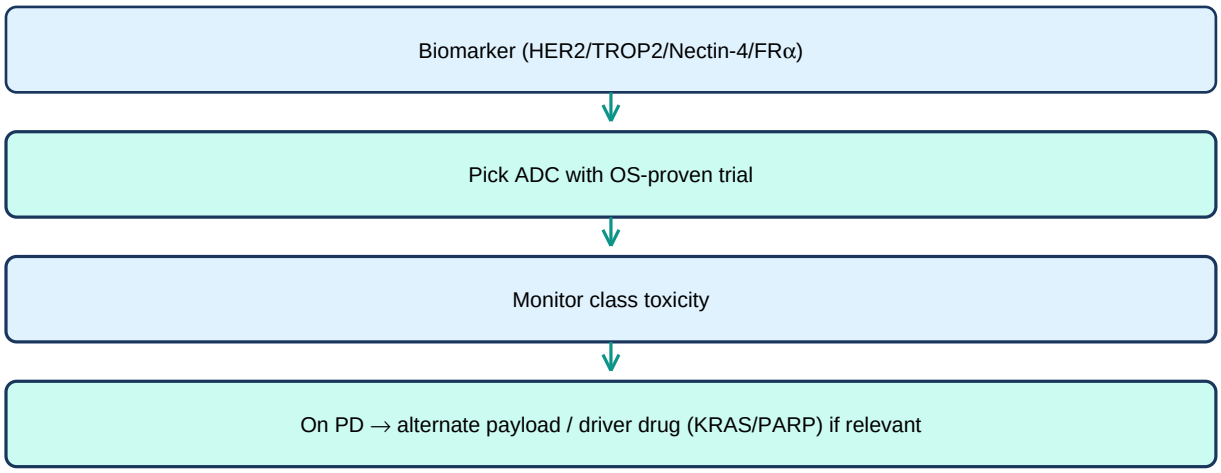


Figure: stepwise exam algorithm

Trial	Key result / exam takeaway
DESTINY program	T-DXd across HER2 spectra
ASCENT / TROPiCS	SG in TNBC & HR+ mBC
EV-302	EV+P 1L urothelial
MIRASOL	Mirvetuximab FRα+ PROC
TROPION / NCCN 2026	Dato-DXd EGFR+ NSCLC subsequent

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
T-DXd	5.4 mg/kg Q3W breast (ILD watch)
SG	10 mg/kg D1,8 Q21D
EV	1.25 mg/kg D1,8,15 Q28D ± pembro
Dato-DXd	Q3W per label — preferred post-osi+chemo

Table: doses and regimens

10. Integrating the triad — shared principles

Biomarker-first oncology: NGS for KRAS allele & HRR/BRCA; IHC/ISH for HER2/FRα/Nectin-4/TROP2 context. Use the agent with proven OS earliest when toxicity allows. Avoid overlapping severe marrow or ILD risks without staggered planning.

Liquid biopsy complements tissue for KRAS resistance and BRCA reversions. Trials remain standard options especially for G12D and post-ADC/post-PARPi resistance.

- One page comparison table: Class | Target | Exemplar drug | Trial | NCCN 2026 note.
- Indian context: access/cost; prioritize Category 1; molecular testing equity.

11. Resistance & sequencing vignettes (examiner favorites)

Vignette A: EGFR NSCLC on osi+chemo → PD → Dato-DXd (ADC), not random chemo. Vignette B: mCRPC BRCA2 → PARPi ± ARPI strategies; consider radioligand later. Vignette C: KRAS G12C CRC progression on sotorasib alone → add/ensure EGFR blockade or switch per evidence. Vignette D: HER2-low mBC after SG → T-DXd if not used (payload switch + antigen).

EXAM PEARLS / HALLMARKS

- Payload switch and allele switch are parallel ideas across ADCs and KRAS.

12. Toxicity comparison table for the three classes

KRAS G12Ci: GI, LFT, ILD rare, QTc (adagrasib). PARPi: marrow, nausea, MDS/AML rare, hypertension (niraparib). ADCs: ILD (DXd), neuropathy (MMAE), ocular (mirv/belantamab), diarrhea (SG).

- Baseline ECG if adagrasib; CNP exams if EV; eye scores if mirvetuximab; CBC q month on PARPi initially.

13. NCCN 2026 master box for the triad

Pull NSCLC (Dato-DXd), ovarian PARP narrowing, prostate niraparib/abi BRCA2, breast ADC/PARP, bladder EV+P into one final box.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* NSCLC v9.2026; Ovarian 2026 updates; Prostate v1.2027; Breast v9.2026; Bladder v3.2026
- Dato-DXd preferred subsequent after osi+chemo in EGFR+ NSCLC.
- PARPi ovarian use narrowed to maintenance/biomarker-defined contexts.
- ADT + niraparib/abiraterone for BRCA2 APMS prostate subsets.
- ADC expansions: T-DXd, SG, EV+P, mirvetuximab, belantamab combos.
- KRAS G12C inhibitors embedded in NSCLC/CRC subsequent pathways with combo nuance in CRC.

14. Summary checklist & pearls

Close: KRAS G12C real (sotorasib/adagrasib), G12D emerging; PARPi across ovary/breast/pancreas/prostate with 2026 ovarian narrowing + prostate BRCA2 combo; ADCs as antigen-directed cytotoxics reshaping NCCN maps.

- G12C ≠ G12D pharmacology
- CRC G12C needs EGFR mAb partner
- OlympiA adjuvant olaparib breast
- EV-302 and DESTINY must-quote ADC trials
- Reversion mutations → PARPi resistance

Triad answer closer

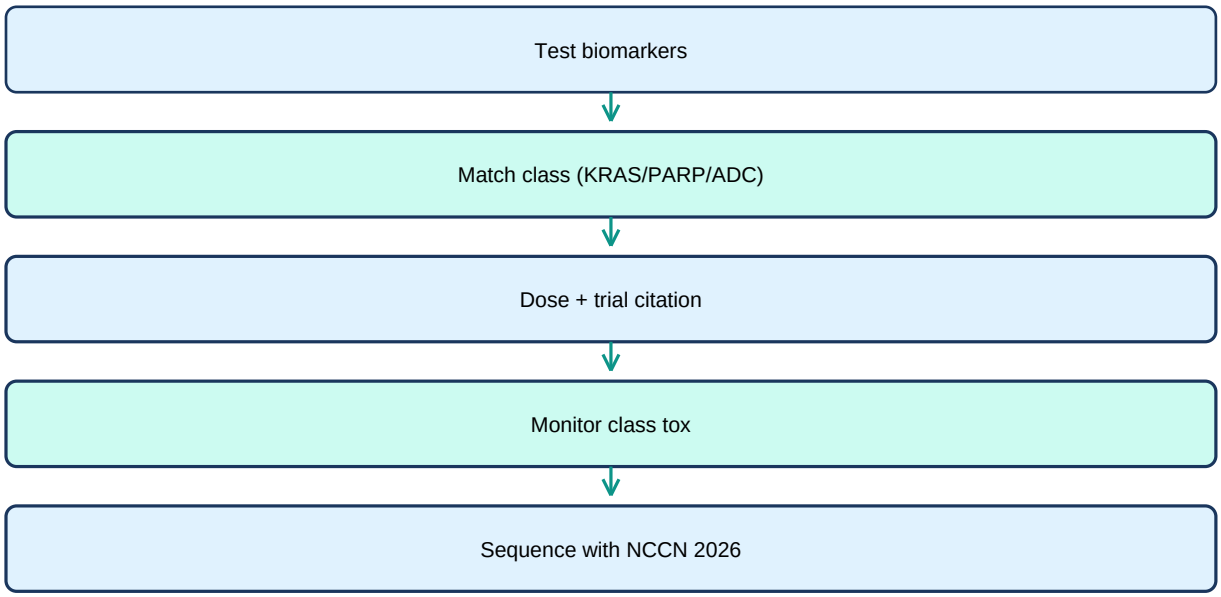


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Three classes, three mechanisms, three NCCN 2026 bullets — minimum passing skeleton.
- Write exact doses for sotorasib 960 mg, adagrasib 600 mg BID, olaparib 300 mg BID.
- Mention synthetic lethality and bystander effect by name.
- Future: pan-RAS, G12D, ADC+IO and PARPi combos carefully toxicity-managed.
- Financial toxicity counseling is fair game in 2026 answers.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

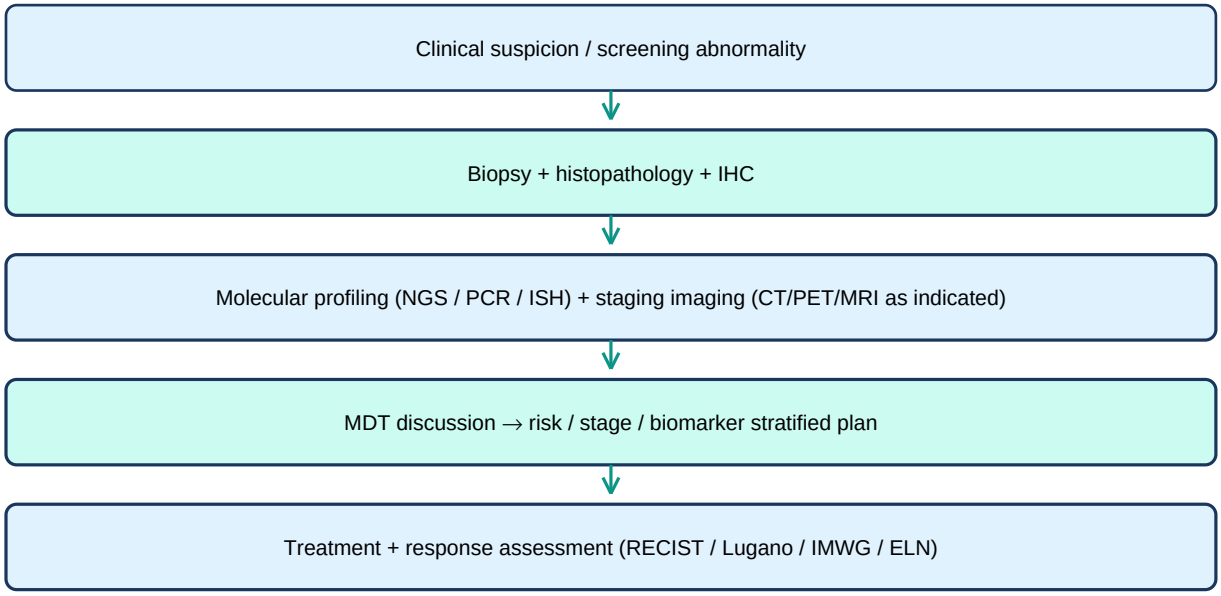


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.

• Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).

• Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.

• Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).

• Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

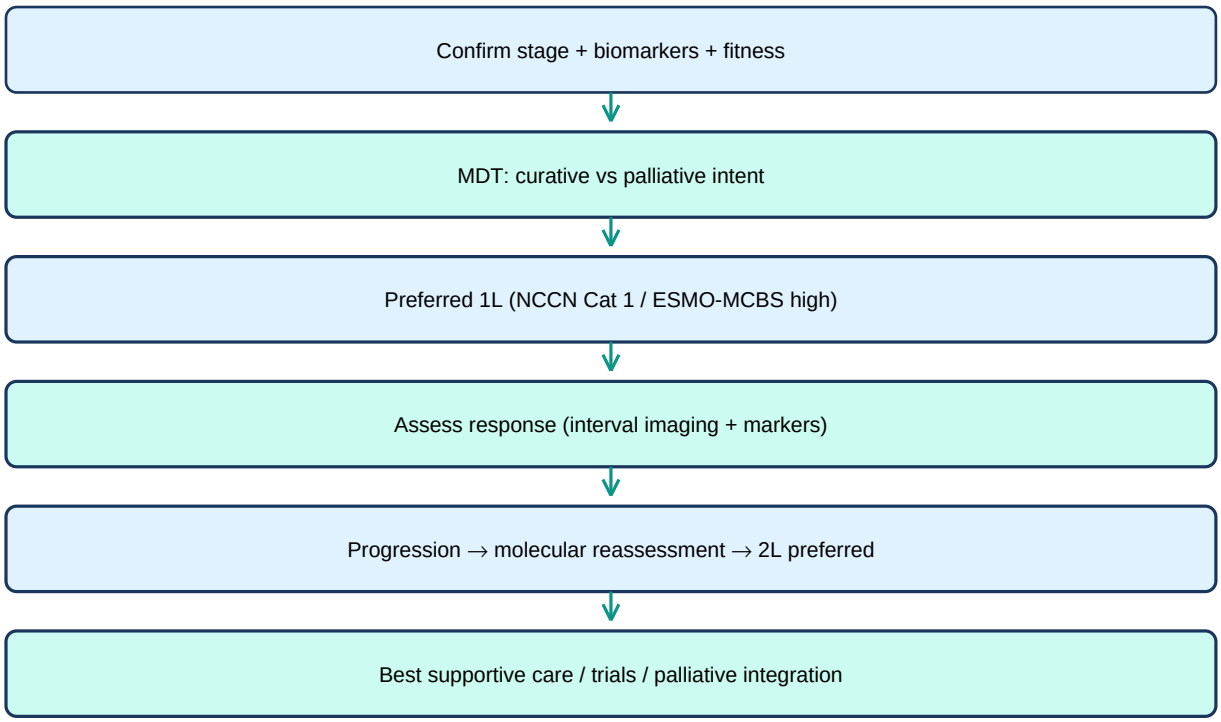


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Niraparib	200–300 mg PO daily (weight/platelets individualized in ovarian)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

• Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.

• Distinguish statistical vs clinical significance; note crossover and subgroup caveats.

• For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
SOLO1 / PAOLA-1 / PRIMA / VELIA	1L PARPi maintenance paradigms
MIRASOL	Mirvetuximab FRα+ platinum-resistant
ROSELLA	Nab-paclitaxel + relacorilant — NCCN 2026 platinum-resistant

KEYNOTE-B96	Paclitaxel + pembro ± bev PD-L1+ OS signal
GOG-172 / iPocc	IP chemo historical; HIPEC OVHIPEC-1 selected
OlympiAD / EMBRACA	Olaparib/talazoparib mBC gBRCA
POLO	Olaparib maintenance pancreatic gBRCA
PROfound	Olaparib mCRPC HRR

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

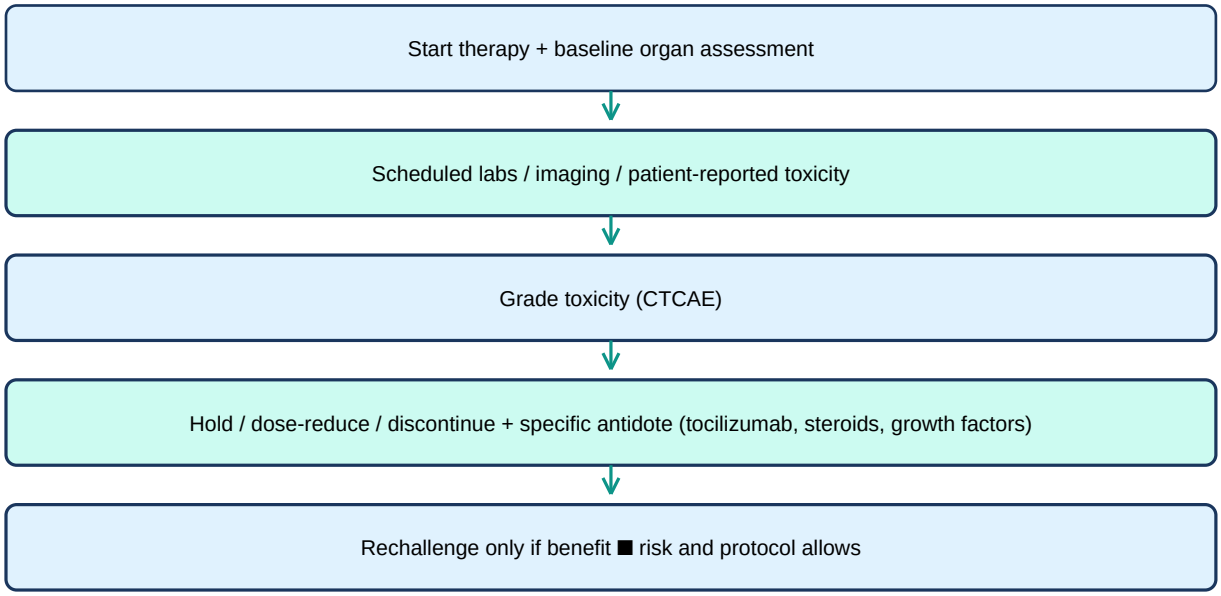


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Non-Small Cell Lung Cancer v9.2026; Small Cell Lung Cancer v1.2027 / updates from v2.2026; Ovarian Cancer (latest 2026 panel updates)
- AJCC 9th edition staging adopted in NCCN NSCLC pathway.

• Osimertinib + platinum/pemetrexed AND amivantamab + lazertinib elevated to Category 1 preferred 1L for EGFR mut+ (detected before 1L systemic therapy).

• Datopotamab deruxtecan (Dato-DXd) preferred 2L after progression on osimertinib + chemotherapy; also preferred 3L after other preferred 1L regimens; listed for EGFR ex19del/L858R, uncommon EGFR, and EGFR ex20ins subsequent therapy.

• Amivantamab + lazertinib useful in certain circumstances after osimertinib progression (multiple lesions) if not previously given.

• Repotrectinib or taletrectinib preferred subsequent for ROS1+ with brain metastases.

• Ado-trastuzumab emtansine useful in certain circumstances for ERBB2/HER2-mutated NSCLC after frontline progression.

• Carboplatin + etoposide + atezolizumab → maintenance lurbinectedin + atezolizumab added for extensive-stage SCLC.

• Consider adding lurbinectedin to maintenance atezolizumab if ≥SD after 4 cycles induction, ECOG 0–1, no prior brain metastases.

• Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

• PARPi narrowed: removed from many recurrent therapy sections; concentrate on maintenance + biomarker-defined (BRCA/HRD).

• Platinum-resistant: nab-paclitaxel + relacorilant preferred (ROSELLA) after ≤3 lines + prior bevacizumab.

• Paclitaxel + pembrolizumab ± bevacizumab useful if PD-L1 CPS ≥1 (KEYNOTE-B96 OS signal).

EXAM PEARLS / HALLMARKS

• NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).

• Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

• MDT documentation improves outcomes and exam completeness.

• Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.

• Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

• Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.

• Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.

• Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.

• Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

• Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)

• Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

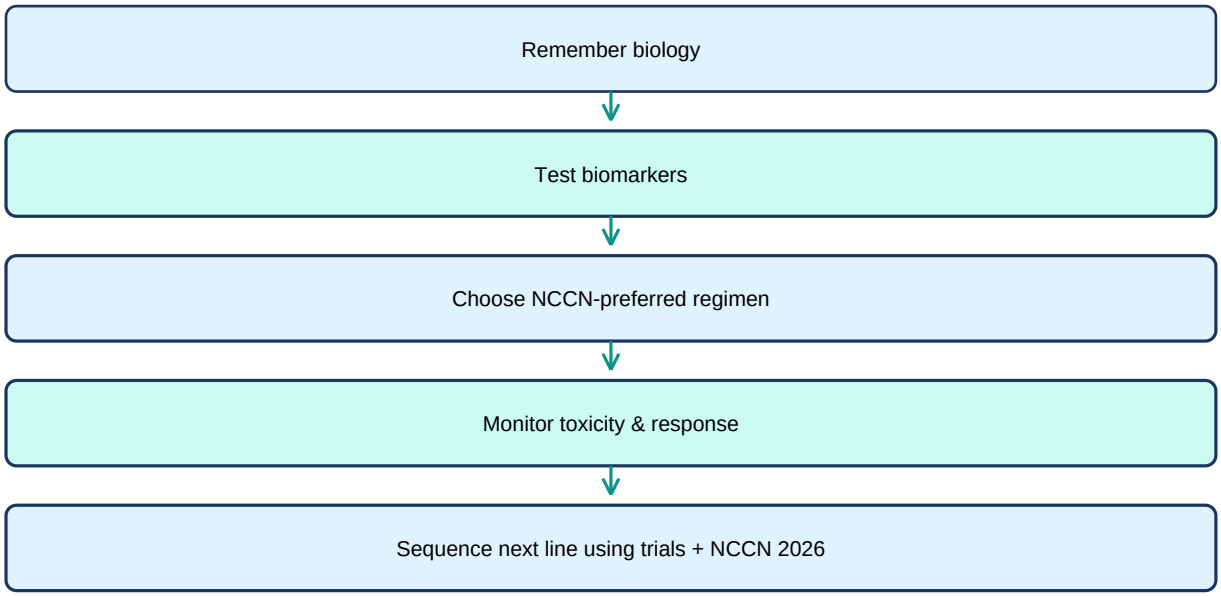


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say ‘NCCN 2026’ explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.

- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Niraparib	200–300 mg PO daily (weight/platelets individualized in ovarian)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.
- ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

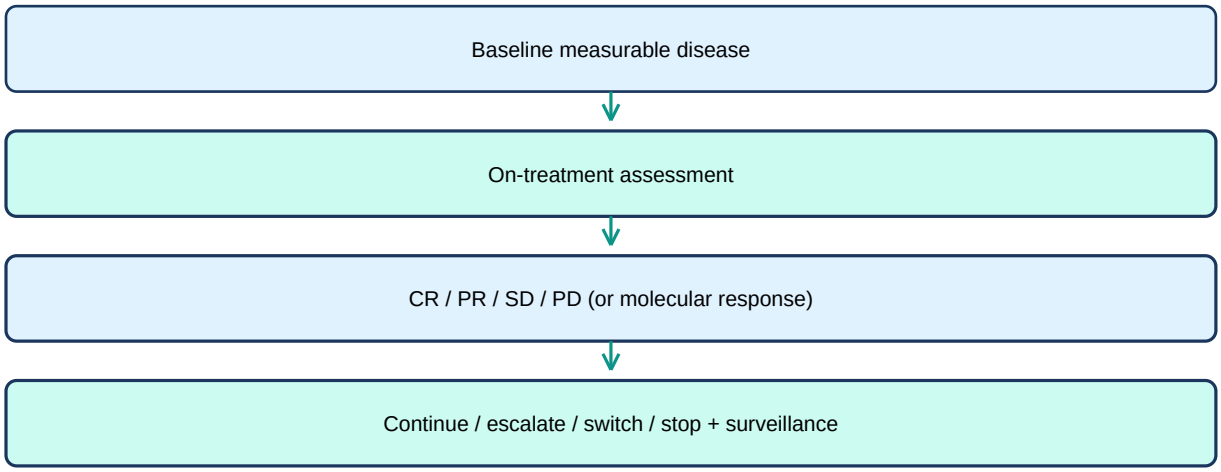


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

- Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.
- Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
SOLO1 / PAOLA-1 / PRIMA / VELIA	1L PARPi maintenance paradigms
MIRASOL	Mirvetuximab FRα+ platinum-resistant
ROSELLA	Nab-paclitaxel + relacorilant — NCCN 2026 platinum-resistant
KEYNOTE-B96	Paclitaxel + pembro ± bev PD-L1+ OS signal
GOG-172 / iPocc	IP chemo historical; HIPEC OVHIPEC-1 selected
OlympiAD / EMBRACA	Olaparib/talazoparib mBC gBRCA

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

- Document MDT decisions; clinical trial referral when standard options exhausted.
- Counsel financial toxicity transparently; prioritize highest-value Category 1 options.
- Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

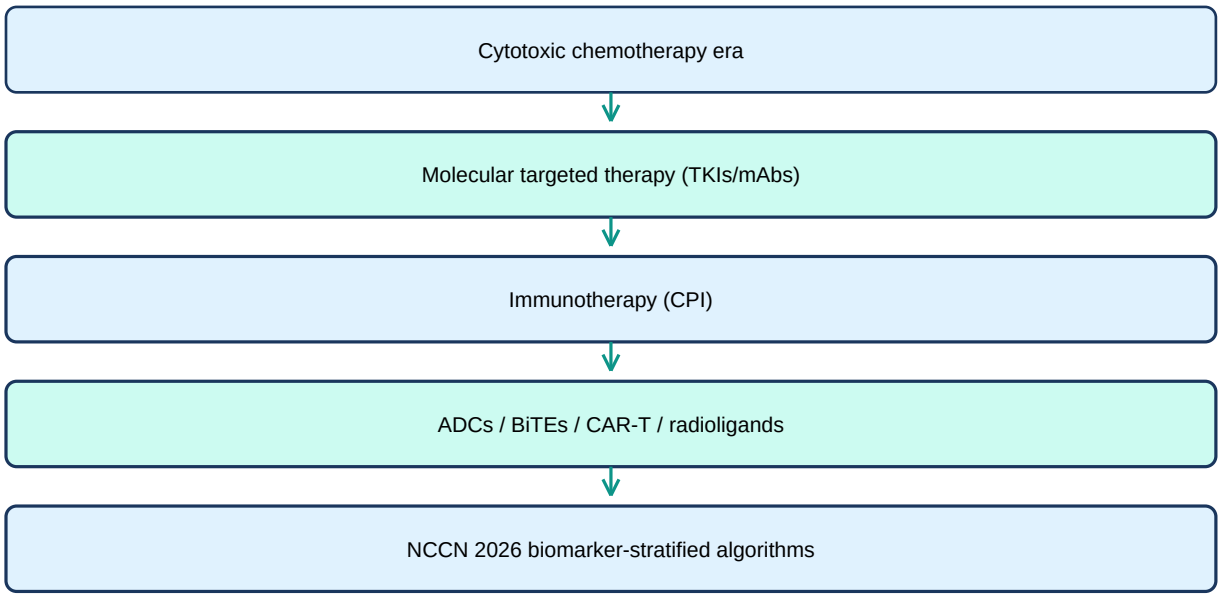


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

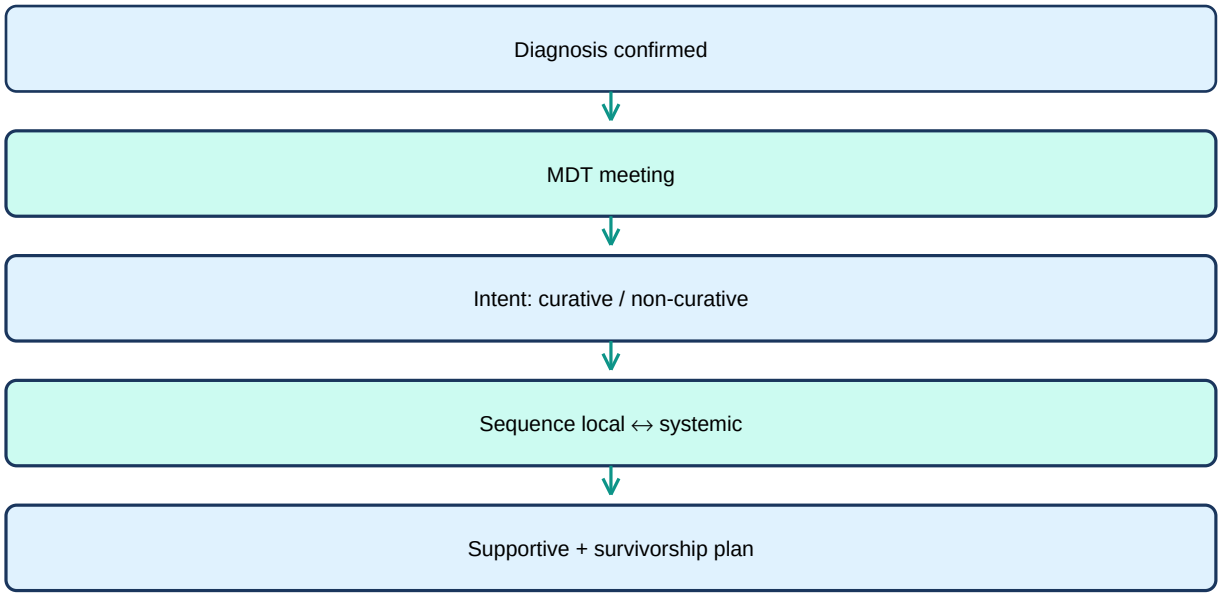


Figure: stepwise exam algorithm

E. PARP inhibitor deep notes

Mechanism: PARP trapping + synthetic lethality in HR-deficient cells. Indications span ovarian maintenance, breast (gBRCA), pancreas (POLO), prostate (HRR), and evolving combinations with ARPI.

NCCN 2026 ovarian: PARPi narrowed in recurrent treatment sections; focus maintenance + biomarkers. Prostate: ADT+niraparib/abiraterone for BRCA2 APMS subsets.

Toxicities: hematologic, GI, fatigue; rare MDS/AML — counsel. Resistance: reversion mutations restoring HR.

Drug / regimen	Dose / schedule (exam pearls)
Olaparib	300 mg BID tablets
Niraparib	200–300 mg daily individualized
Talazoparib	1 mg daily (breast gBRCA)

Table: doses and regimens

E. Antibody–drug conjugate deep notes

ADC anatomy: mAb + linker + payload. Cleavable linkers + permeable payloads → bystander effect (T-DXd). Non-cleavable (T-DM1) more intracellularly restricted.

Payloads: microtubule inhibitors (DM1, MMAE) vs topoisomerase-I (DXd, SN-38).

NCCN 2026 expands ADCs across tumors: T-DXd, Dato-DXd, SG, EV, mirvetuximab, belantamab combos, loncastuximab.

mAb	Targets HER2/TROP2/Nectin-4/FRα/BCMA/CD30
Linker	Cleavable vs non-cleavable
Payload	Topo-I or microtubule toxin
Bystander	Kills neighboring low-antigen cells
Key toxicity	ILD / neuropathy / hyperglycemia / ocular

Figure: mechanism / pathway map

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).

- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

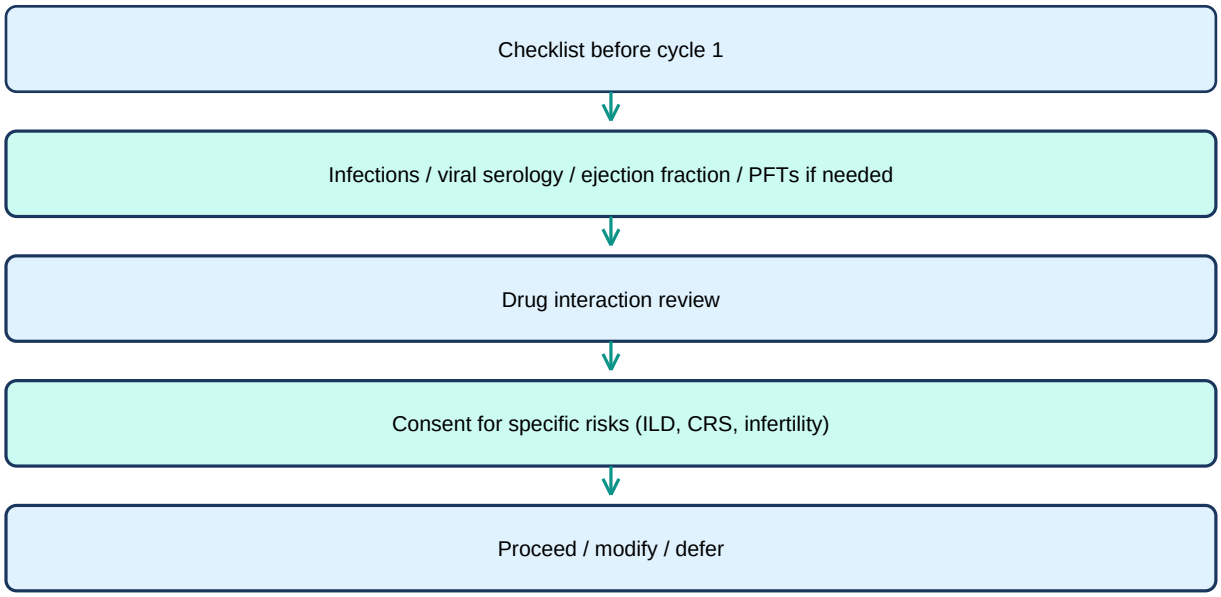


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

• Wall chart + flowchart + trial table = visual completeness.

• Write 'NCCN 2026' in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
SOLO1 / PAOLA-1 / PRIMA / VELIA	1L PARPi maintenance paradigms
MIRASOL	Mirvetuximab FRα+ platinum-resistant
ROSELLA	Nab-paclitaxel + relacorilant — NCCN 2026 platinum-resistant
KEYNOTE-B96	Paclitaxel + pembro ± bev PD-L1+ OS signal
GOG-172 / iPocc	IP chemo historical; HIPEC OVHIPEC-1 selected
OlympiAD / EMBRACA	Olaparib/talazoparib mBC gBRCA
POLO	Olaparib maintenance pancreatic gBRCA
PROfound	Olaparib mCRPC HRR

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Niraparib	200–300 mg PO daily (weight/platelets individualized in ovarian)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Non-Small Cell Lung Cancer v9.2026; Small Cell Lung Cancer v1.2027 / updates from v2.2026; Ovarian Cancer (latest 2026 panel updates)

• AJCC 9th edition staging adopted in NCCN NSCLC pathway.

• Osimertinib + platinum/pemetrexed AND amivantamab + lazertinib elevated to Category 1 preferred 1L for EGFR mut+ (detected before 1L systemic therapy).

• Datopotamab deruxtecan (Dato-DXd) preferred 2L after progression on osimertinib + chemotherapy; also preferred 3L after other preferred 1L regimens; listed for EGFR ex19del/L858R, uncommon EGFR, and EGFR ex20ins subsequent therapy.

• Amivantamab + lazertinib useful in certain circumstances after osimertinib progression (multiple lesions) if not previously given.

• Repotrectinib or taletrectinib preferred subsequent for ROS1+ with brain metastases.

• Ado-trastuzumab emtansine useful in certain circumstances for ERBB2/HER2-mutated NSCLC after frontline progression.

• Carboplatin + etoposide + atezolizumab → maintenance lurbinectedin + atezolizumab added for extensive-stage SCLC.

• Consider adding lurbinectedin to maintenance atezolizumab if ≥SD after 4 cycles induction, ECOG 0–1, no prior brain metastases.

• Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

• PARPi narrowed: removed from many recurrent therapy sections; concentrate on maintenance + biomarker-defined (BRCA/HRD).

• Platinum-resistant: nab-paclitaxel + relacorilant preferred (ROSELLA) after ≤3 lines + prior bevacizumab.

• Paclitaxel + pembrolizumab ± bevacizumab useful if PD-L1 CPS ≥1 (KEYNOTE-B96 OS signal).

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn

- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Non-Small Cell Lung Cancer v9.2026; Small Cell Lung Cancer v1.2027 / updates from v2.2026; Ovarian Cancer (latest 2026 panel updates)
- AJCC 9th edition staging adopted in NCCN NSCLC pathway.
- Osimertinib + platinum/pemetrexed AND amivantamab + lazertinib elevated to Category 1 preferred 1L for EGFR mut+ (detected before 1L systemic therapy).
- Datopotamab deruxtecan (Dato-DXd) preferred 2L after progression on osimertinib + chemotherapy; also preferred 3L after other preferred 1L regimens; listed for EGFR ex19del/L858R, uncommon EGFR, and EGFR ex20ins subsequent therapy.
- Amivantamab + lazertinib useful in certain circumstances after osimertinib progression (multiple lesions) if not previously given.
- Repotrectinib or taletrectinib preferred subsequent for ROS1+ with brain metastases.
- Ado-trastuzumab emtansine useful in certain circumstances for ERBB2/HER2-mutated NSCLC after frontline progression.
- Carboplatin + etoposide + atezolizumab → maintenance lurbinectedin + atezolizumab added for extensive-stage SCLC.
- Consider adding lurbinectedin to maintenance atezolizumab if ≥SD after 4 cycles induction, ECOG 0–1, no prior brain metastases.
- Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).
- PARPi narrowed: removed from many recurrent therapy sections; concentrate on maintenance + biomarker-defined (BRCA/HRD).
- Platinum-resistant: nab-paclitaxel + relacorilant preferred (ROSELLA) after ≤3 lines + prior bevacizumab.
- Paclitaxel + pembrolizumab ± bevacizumab useful if PD-L1 CPS ≥1 (KEYNOTE-B96 OS signal).

Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Non-Small Cell Lung Cancer v9.2026; Small Cell Lung Cancer v1.2027 / updates from v2.2026; Ovarian Cancer (latest 2026 panel updates)
- AJCC 9th edition staging adopted in NCCN NSCLC pathway.
- Osimertinib + platinum/pemetrexed AND amivantamab + lazertinib elevated to Category 1 preferred 1L for EGFR mut+ (detected before 1L systemic therapy).
- Datopotamab deruxtecan (Dato-DXd) preferred 2L after progression on osimertinib + chemotherapy; also preferred 3L after other preferred 1L regimens; listed for EGFR ex19del/L858R, uncommon EGFR, and EGFR ex20ins subsequent therapy.
- Amivantamab + lazertinib useful in certain circumstances after osimertinib progression (multiple lesions) if not previously given.
- Repotrectinib or taletrectinib preferred subsequent for ROS1+ with brain metastases.
- Ado-trastuzumab emtansine useful in certain circumstances for ERBB2/HER2-mutated NSCLC after frontline progression.
- Carboplatin + etoposide + atezolizumab → maintenance lurbinectedin + atezolizumab added for extensive-stage SCLC.
- Consider adding lurbinectedin to maintenance atezolizumab if ≥SD after 4 cycles induction, ECOG 0–1, no prior brain metastases.

- Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).
- PARPi narrowed: removed from many recurrent therapy sections; concentrate on maintenance + biomarker-defined (BRCA/HRD).
- Platinum-resistant: nab-paclitaxel + relacorilant preferred (ROSELLA) after ≤3 lines + prior bevacizumab.
- Paclitaxel + pembrolizumab ± bevacizumab useful if PD-L1 CPS ≥1 (KEYNOTE-B96 OS signal).

Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.